SAMAGRA SHIKSHA KERALA — SECOND TERM SUMMATIVE ASSESSMENT

STANDARD IX | PHYSICS | OFFICIAL ENGLISH ANSWER KEY & SOLUTIONS

Standard: Class 9 • Medium: English • Max Marks: 40 • Time: 1.5 Hours

SECTION A — Objective & Multiple Choice Questions (1 Mark Each)

Question 1 (1 Mark)

Q: When a ray of light enters obliquely from air into water, the angle of incidence is 50°. Then the angle of refraction will be:
(a) greater than 50° with the normal (b) equal to 50° with the normal
(c) equal to 90° with the normal (d) less than 50° with the normal

Answer & Explanation:

Correct Option: (d) less than 50° with the normal
Reason / Principle: Water is optically denser than air ($n_{water} = 1.33 > n_{air} = 1.00$). When a light ray travels obliquely from an optically rarer medium (air) to an optically denser medium (water), it bends towards the normal. Consequently, the angle of refraction (r) is strictly less than the angle of incidence ($i = 50°$).

Question 2 (1 Mark)

Q: Statement: When an object moves along a circular path with uniform speed, its acceleration is directed away from the centre.
Reason: The direction of velocity changes at every point on the circular path.

Answer & Explanation:

Correct Option: (c) Statement is incorrect, reason is correct.
Explanation: For an object moving with uniform speed along a circular path, the acceleration is the centripetal acceleration, which is always directed **towards the centre** of the circle (not away from it). Hence, the Statement is incorrect. However, velocity is a vector quantity; along a curved path, the direction of motion continuously changes at every single point. Hence, the Reason is correct.

Question 3 (1 Mark)

Q: Match the following physical quantities with their definitions:
(i) Rate of change of momentum (x) Velocity
(ii) Change in momentum (y) Force
(iii) Rate of change of displacement (z) Impulse

Answer & Explanation:

Correct Option: (a) i – y, ii – z, iii – x
• **Rate of change of momentum** = Force ($F = \Delta p / \Delta t$)
• **Change in momentum** = Impulse ($J = F \cdot \Delta t = \Delta p$)
• **Rate of change of displacement** = Velocity ($v = \Delta s / \Delta t$)

Question 4 (1 Mark)

Q: Identify the situation(s) where work done is positive:
(a) A mango falling from a tree (b) Applying brakes to stop a moving car
(c) A bucket submerged in water is being lifted upward

Answer & Explanation:

Correct Situations: (a) and (c)
• **(a) A mango falling from a tree:** The downward gravitational force and the downward displacement are in the same direction ($\theta = 0°$, $\cos 0° = +1$), so work done by gravity is positive ($W > 0$).
• **(c) Lifting a bucket from water:** The upward applied lifting force and upward displacement are in the same direction, so work done by the lifting force is positive.
Note: In (b), braking friction opposes forward motion ($\theta = 180°$), producing negative work.

SECTION B — Short Answer Questions (2 Marks Each)

Question 5 (2 Marks)

Q: Two persons push a box in the same direction with a force of 500 N each.
(a) Find the total force applied. (b) Is it a balanced force or unbalanced force?

Answer & Explanation:

(a) Total force applied:

Since both persons push in the same direction, their forces add algebraically:
$F_{total} = F_1 + F_2 = 500\ N + 500\ N = 1000\ N$ (in the direction of the push).
(b) Nature of force:

It is an **unbalanced force**. An unbalanced force exists when the vector sum (net resultant) of all forces acting on an object is non-zero ($F_{net} \neq 0$), causing an acceleration in the direction of the net force.

Question 6 (A or B) (2 Marks)

Q: From the given velocity-time graph (OA: 0 to 100 m/s in 2 s; AB: uniform 100 m/s from 2 to 5 s; BC: 100 to 0 m/s from 5 to 7 s):
[A] (a) What is the acceleration during stage AB? (b) What does stage BC indicate?
— OR — [B] Calculate the displacement of the object during the period of uniform velocity.

Answer & Explanation:

[Option A]

(a) Acceleration during stage AB: During stage AB (time $t = 2$ s to $t = 5$ s), the object maintains a constant speed of 100 m/s. Therefore, **Acceleration = 0 m/s²** (uniform velocity).

(b) Significance of stage BC: Stage BC represents **uniform retardation (negative acceleration / deceleration)**, where the velocity decreases steadily from 100 m/s to 0 m/s.

— OR — [Option B]

Displacement during uniform velocity (Stage AB):
Time interval $\Delta t = 5\ s - 2\ s = 3\ s$. Constant velocity $v = 100$ m/s.
Displacement s = velocity × time = 100 m/s × 3 s = 300 m.
(Alternatively, Area under rectangle AB = 100 × 3 = 300 m).

Question 7 (2 Marks)

Q: An object moves with uniform speed along a curved path. Is the velocity of the object uniform? Justify your answer.

Answer & Explanation:

Answer: No, the velocity of the object is NOT uniform.
Justification: Velocity is a vector quantity that depends on both magnitude (speed) and direction of motion. Even though the magnitude of velocity (speed) remains constant, the direction of motion continuously changes at every point along a curved path. Since the direction changes continuously, the velocity is non-uniform and the object undergoes acceleration.

SECTION C — Conceptual & Analytical Questions (3 Marks Each)

Question 8 (3 Marks)

Q: Give scientific reasons:
(a) Passengers in a moving vehicle are advised to wear seat belts.
(b) A cricketer moves his hands backward while catching a fast cricket ball.

Answer & Explanation:

(a) Why passengers wear seat belts:

Based on **Inertia of Motion (Newton's First Law)**. When a vehicle moves at high speed, the passengers share that forward velocity. In the event of sudden braking or collision, the passenger's lower body is brought to rest with the vehicle, but their upper body tends to keep moving forward due to inertia. Seat belts apply a safe restraining force, preventing passengers from striking the dashboard or windshield.

(b) Why a cricketer pulls hands backward while catching:

Based on Newton's Second Law of Motion ($F = \Delta p / \Delta t$). By pulling hands backward, the cricketer increases the time interval (Δt) taken to bring the momentum of the ball to zero. Since force is inversely proportional to time, a longer time significantly reduces the rate of change of momentum, reducing the impact force on the hands and preventing injury.

Question 9 (A or B) (3 Marks)

Q: [A] A force of 500 N acts on a body of mass 100 kg at rest. It moves through a distance of 16 m. Calculate the work done.
— OR — [B] A body of mass 4 kg moving with a velocity of 20 m/s is brought to rest by a retarding force. Find the work done.

Answer & Explanation:

[Option A]

Given: Mass m = 100 kg, Applied Force F = 500 N, Displacement s = 16 m.
Formula: Work Done W = Force × Displacement = $F \cdot s$
W = 500 N × 16 m = 8000 J = 8 kJ.
(Alternatively: Acceleration $a = F/m = 500/100 = 5$ m/s²; $W = m \cdot a \cdot s = 100 \times 5 \times 16 = 8000$ J).

— OR — [Option B]

Given: Mass m = 4 kg, Initial velocity u = 20 m/s, Final velocity v = 0 m/s.
According to the Work-Energy Theorem, Work Done = Change in Kinetic Energy (ΔK):
$W = K_{final} - K_{initial} = 0 - \frac{1}{2} m u^2$
$W = -\frac{1}{2} \times 4 \text{ kg} \times (20 \text{ m/s})^2 = -\frac{1}{2} \times 4 \times 400 = -800$ J.
The work done by the retarding force is –800 J (Magnitude = 800 Joules).

Question 10 (3 Marks)

Q: (a) What is meant by free fall?
(b) Why does a freely falling body experience weightlessness?

Answer & Explanation:

(a) Definition of Free Fall:

Free fall is the motion of a body falling towards the Earth solely under the influence of the Earth's gravitational pull, with no other external opposing forces (such as air resistance) acting upon it.

(b) Explanation of Weightlessness:

Weight felt by a body is the normal reaction force (R) exerted on it by the supporting surface. When a body falls freely, its downward acceleration equals acceleration due to gravity ($a = g$).
The apparent weight is given by: $R = m(g - a) = m(g - g) = 0$.
Since the supporting surface exerts no reaction force on the body, the body experiences a state of complete weightlessness.

Question 11 (3 Marks)

Q: (a) What is centripetal force?
(b) If the centripetal force suddenly ceases to act on a body in circular motion, what will be the direction of its subsequent motion?

Answer & Explanation:

(a) Centripetal Force:

Centripetal force is the net inward force required to keep an object moving along a circular path with uniform speed. It acts continuously along the radius directed towards the centre of the circular path. Formula: $F_c = m\,v^2 / r$.

(b) Direction of motion when centripetal force ceases:

If the centripetal force suddenly ceases to act, the object will fly off **tangentially along a straight line** in the direction of its instantaneous velocity at that exact instant, due to **inertia of direction** (Newton's First Law).

Question 12 (3 Marks)

Q: Refractive indices: Water = 1.33, Kerosene = 1.44, Turpentine = 1.47, Crown Glass = 1.52.
(a) In which medium does light travel with maximum speed? Why?
(b) A ray of light passes from turpentine into glycerol (refractive index 1.47) at an angle of incidence of 30°. What will be the angle of refraction? Explain.

Answer & Explanation:

(a) Medium with maximum speed of light:

Water. Speed of light in a medium is given by $v = c / n$, where c is the speed of light in vacuum and n is the refractive index. Since refractive index is inversely proportional to speed, the medium with the lowest refractive index has the highest optical speed. Among the given media, Water has the lowest refractive index ($n = 1.33$).

(b) Angle of refraction from turpentine into glycerol:

Angle of refraction $r = 30°$.
Reason: Refractive index of turpentine = 1.47 and refractive index of glycerol = 1.47. Since both media have identical refractive indices, their optical densities are exactly equal. Therefore, light does not undergo any change in speed or deviation at the boundary. The ray passes undeviated through the interface, keeping $r = i = 30°$.

Question 13 (3 Marks)

Q: Identify which law of motion explains each of the following observations:
(a) Recoil of a gun when a bullet is fired.
(b) Dust particles fall off when a carpet is beaten with a stick.
(c) High-jump athletes land on thick foam cushions or sand pits.

Answer & Explanation:

(a) Recoil of a gun when a bullet is fired:

Newton's Third Law of Motion (To every action there is an equal and opposite reaction) & **Law of Conservation of Linear Momentum**. The forward force on the bullet produces an equal backward recoil force on the gun.

(b) Dust falls off when a carpet is beaten with a stick:

Newton's First Law of Motion (Inertia of Rest). When struck, the carpet moves forward suddenly, but the dust particles tend to remain in their state of rest due to inertia and separate.

(c) High-jump athletes landing on foam cushions or sand:

Newton's Second Law of Motion (Impulse-Momentum Principle). The soft surface increases the deceleration time interval (Δt), which minimizes the rate of change of momentum and drastically reduces the impact force ($F = \Delta p / \Delta t$).

SECTION D — Comprehensive Analytical & Numerical Problems (4 Marks Each)

Question 14 (A or B) (4 Marks)

Q: [A] Mass of an object on Earth is 18 kg (g = 10 m/s², g_moon = g/6):
(a) What is its weight on Earth? (b) What will be its mass and weight on the Moon?
— OR — [B] (a) What is the weight of an object at the centre of the Earth?
(b) State two factors on which weight depends. (c) Where on Earth's surface is weight maximum? Why?

Answer & Explanation:

[Option A]

Given: Mass on Earth m = 18 kg, g_{earth} = 10 m/s², g_{moon} = g_{earth} / 6.
(a) Weight on Earth:
$W_{Earth} = m \cdot g_{earth}$ = 18 kg × 10 m/s² = 180 N.
(b) Mass and Weight on Moon:
- **Mass on Moon = 18 kg (Mass is an intrinsic scalar property that never changes with location).**
- **Weight on Moon: $W_{Moon} = W_{Earth}$ / 6 = 180 N / 6 = 30 N.**

— OR — [Option B]

(a) Weight of an object at the centre of the Earth: 0 N (Zero), because gravitational acceleration at Earth's centre is zero (g = 0).
(b) Two factors determining weight: 1. Mass of the body (m); 2. Local acceleration due to gravity (g), since $W = m \cdot g$.
(c) Location of maximum weight: At the **Poles**. Earth is an oblate ellipsoid with a smaller polar radius than equatorial radius ($R_{pole} < R_{equator}$). Since $g = G \cdot M / R^2$, g is inversely proportional to R^2, making g maximum at the poles. Therefore, W is maximum at the poles.

Question 15 (A or B) (4 Marks)

Q: [A] An object weighs 0.45 kgwt in air. When completely immersed in water, it weighs 0.31 kgwt.
(a) Find the loss of weight. (b) What is the buoyant force? (c) Find weight of water displaced.
— OR — [B] A ship sails from sea water into river water.
(a) How does buoyant force change? Does it sink deeper or rise? Explain. (b) What determines buoyant force?

Answer & Explanation:

[Option A]

Given: Weight in air W_{air} = 0.45 kgwt, Weight in water W_{water} = 0.31 kgwt.
(a) Loss of weight: $W_{air} - W_{water}$ = 0.45 – 0.31 = 0.14 kgwt (= 0.14 × 9.8 N = 1.372 N).
(b) Buoyant force: By Archimedes' Principle, Buoyant Force = Loss in weight = 0.14 kgwt (= 1.372 N).
(c) Weight of water displaced: Weight of displaced water = Buoyant Force = 0.14 kgwt (= 1.372 N).

— OR — [Option B]

(a) Ship sailing from sea into river water:
• **Buoyant force remains UNCHANGED:** For any freely floating object, the buoyant force always equals the total weight of the ship ($F_b = W_{ship}$).
• **Ship sinks DEEPER:** River water has lower density than sea water ($\rho_{river} < \rho_{sea}$). Since $F_b = V_{sub} \cdot \rho \cdot g$, to displace the same weight in less dense water, the volume of water displaced must increase. Hence, the ship sinks deeper into the river.
(b) Factors determining buoyant force: 1. Volume of the submerged portion of the object (V); 2. Density of the liquid (ρ).

Question 16 (4 Marks)

Q: (a) Name the instrument used to check the purity of milk.
(b) On which scientific principle does this instrument work?
(c) Name another instrument based on the same principle used to measure liquid density.

Answer & Explanation:

(a) Instrument to check purity of milk:

Lactometer (a specially calibrated hydrometer for milk).

(b) Scientific principle:

Principle of Floatation / Archimedes' Principle. When a body floats in a liquid, the weight of the floating body is equal to the weight of the liquid displaced by its submerged portion.

(c) Another instrument based on the same principle:

Hydrometer (measures relative density of liquids; other examples: Alcoholometer, Acid hydrometer for battery testing).

Question 17 (4 Marks)

Q: A person lifts a load of mass 15 kg to a vertical height of 1.5 m. Calculate the work done against gravity. (Take g = 10 m/s²).

Answer & Explanation:

Given:

Mass m = 15 kg, Height h = 1.5 m, Acceleration due to gravity g = 10 m/s².
Formula: Work done against gravity = Potential Energy gained = $W = m \cdot g \cdot h$
Calculation:
W = 15 kg × 10 m/s² × 1.5 m = 225 Joules (J).
Final Answer: The work done against gravity is 225 J.

Question 18 (A or B) (4 Marks)

Q: [A] A ball of mass 0.5 kg moving with a velocity of 65 m/s is brought down to a velocity of 5 m/s by a retarding force of 20 N. Calculate the time taken.
— OR — [B] A force of 14 N acts on a body of mass 0.7 kg moving initially at 30 m/s for 2 seconds in the direction of motion.
(a) Find the change in momentum. (b) Calculate the final velocity of the body.

Answer & Explanation:

[Option A]

Given: Mass m = 0.5 kg, Initial velocity u = 65 m/s, Final velocity v = 5 m/s, Retarding force F = –20 N.
1. Acceleration: $a = F / m$ = –20 N / 0.5 kg = –40 m/s²
2. Using kinematics equation: $v = u + a \cdot t$
$5 = 65 + (-40)t$
$40t = 65 - 5 = 60$
$t = 60 / 40 = 1.5$ seconds.
Final Answer: The time taken to reduce the velocity is 1.5 seconds.

— OR — [Option B]

Given: Mass m = 0.7 kg, Initial velocity u = 30 m/s, Applied Force F = 14 N, Duration t = 2 s.
(a) Change in momentum (Δp):
By Impulse-Momentum Theorem: Δp = Force × Time = $F \cdot t$
Δp = 14 N × 2 s = 28 kg·m/s (or N·s).
(b) Final velocity (v):
$\Delta p = m(v - u)$
$28 = 0.7(v - 30) \Rightarrow v - 30 = 28 / 0.7 = 40$
$v = 30 + 40 = 70$ m/s.
Final Answer: Change in momentum is 28 kg·m/s and final velocity is 70 m/s.